

Course Code	AAI101/AAI201
Title of the Course	Introduction to AI and Prompt Engineering
Course Category	Engineering Science Course (ESC)
Credits	02
NCrF Hours	30
Tools Used	ChatGPT / Claude / Gemini / Copilot (free tier) — browser based

Main Focus of the Course Outcomes

Employability	Skill Development	Entrepreneurship
✓	✓	✓

Course Objectives:

1. Understand what Artificial Intelligence is, its history, and how it is changing every engineering field
2. Understand key AI concepts — Machine Learning, Deep Learning, and Neural Networks — at a conceptual level
3. Explore how AI is applied across domains — healthcare, agriculture, manufacturing, finance, and education
4. Understand Generative AI — how large language models (LLMs) like ChatGPT and Gemini work
5. Learn and practise Prompt Engineering — the skill of communicating effectively with AI tools to get useful results

Course Content

Unit	Topics	Weightage
1.	Introduction to Artificial Intelligence 1. Definition and scope of AI; brief history — from Alan Turing to ChatGPT	20%

	1. Types of AI — Narrow AI, General AI, Super AI (concept) 2. AI vs Machine Learning vs Deep Learning — what is the difference 3. How machines learn — data, patterns, and predictions (concept, no maths) 4. AI today — where it already exists in daily life (Google Maps, Spotify, Netflix, Google Photos, voice assistants) 5. Machine Learning Concepts, Learning Strategies- Supervised, Unsupervised, Reinforcement Learning 6. Data, Data Visualization, Data Pre-processing Class Activity: Students list 5 AI tools they use every day and discuss how each one 'learns'. <i>Branch connect: CE — AI in structural health monitoring; ME — AI in predictive maintenance; EE — smart grid; CSE — AI in compilers; BT — drug discovery; AG — crop yield prediction.</i>	
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How Machines Learn — Data, Patterns, and Predictions

DATA => Find Patterns => LEARN => Create Model => New Data => PREDICTION

Machine learning allows a computer to learn from data and make predictions or decisions without being given every rule manually.

1. Data

Data is the information given to a machine for learning.

Examples:

- Pictures of cats and dogs
- Student marks and attendance
- Previous house prices
- Emails marked as spam or not spam

2. Patterns

A pattern is something that repeatedly appears in the data.

For example, if we give a machine many pictures of cats and dogs:

- Cats may have certain shapes, ears, eyes, etc.
- Dogs may have different shapes and features.

3. Learning

The machine studies the data and the patterns found in it.

It creates a model that represents what it has learned.

For example:

Data → Find Patterns → Learn → Create Model

The model can then be used on new data.

4. Prediction

After learning, the machine receives new data that it has not seen before.

It uses the patterns it learned to make a prediction.

Example:

If a machine has learned from thousands of cat and dog images, we give it a new image.

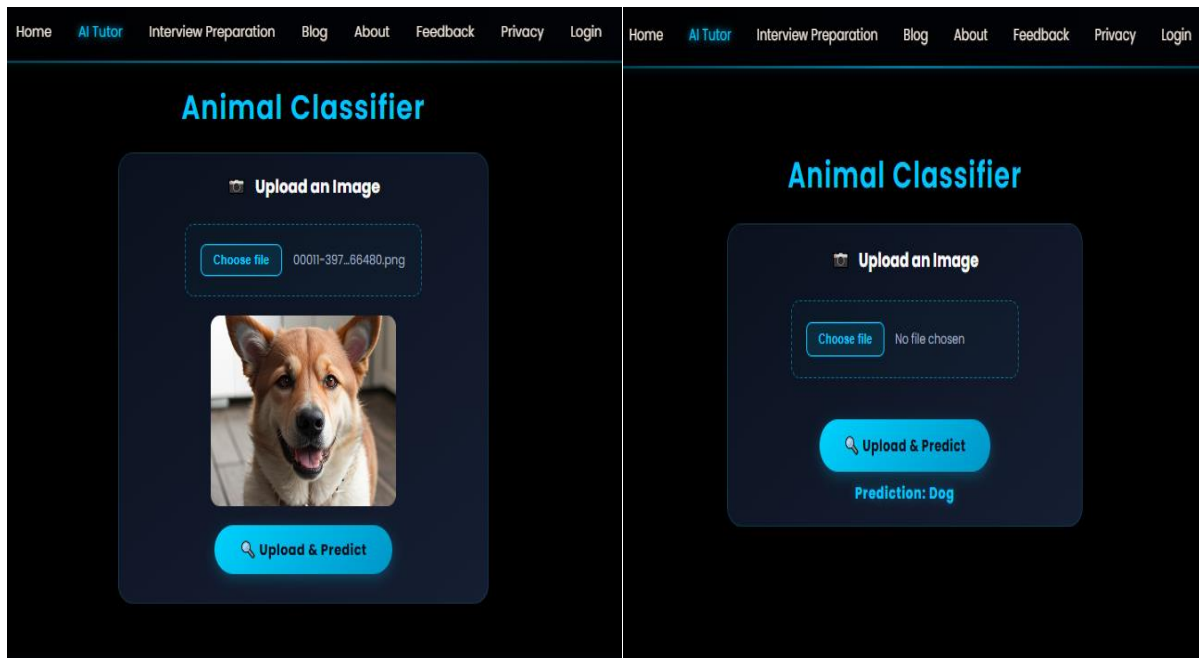
It may predict:

"This is a Cat."

5. Example

Suppose we want to predict whether a student will pass or fail.

Step	What happens
Data	Previous students' attendance, study hours, and marks are given.
Patterns	Machine finds that students with good attendance and regular study often pass.
Learning	Machine learns these patterns.
New Data	Information about a new student is given.
Prediction	Machine predicts whether the student is likely to pass or fail.



AI Today — Where AI Exists in Daily Life

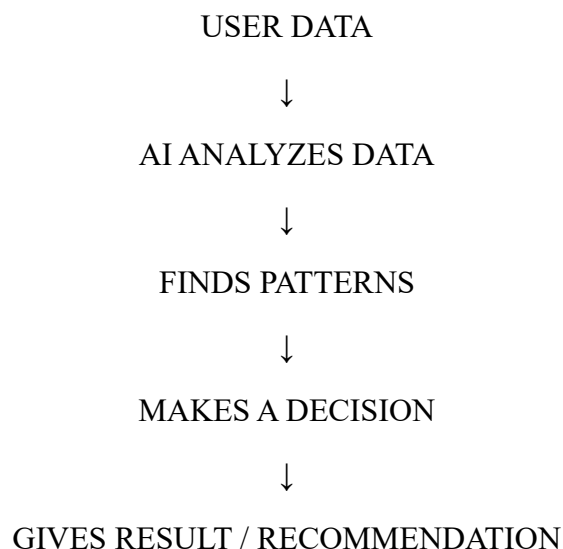
Artificial Intelligence (AI) is not only used in robots or advanced machines. **AI is already a part of our everyday life.** Many apps and services use AI to understand our behavior, recognize information, make recommendations, and help us make decisions.

No.	Daily-Life Example	How AI Is Used	Simple Example
1	Google Maps	AI analyzes traffic, routes, and travel information.	It suggests the fastest route to your destination.
2	Spotify	AI studies your listening habits and recommends music.	It suggests songs similar to the songs you usually listen to.
3	Netflix	AI analyzes your watching history and preferences.	It recommends movies and shows you may like.
4	Google Photos	AI recognizes faces, objects, places, and images.	It can group photos of the same person automatically.
5	Voice Assistants	AI understands human speech and responds to commands.	Siri, Google Assistant, and Alexa can answer questions or perform tasks.

No.	Daily-Life Example	How AI Is Used	Simple Example
6	YouTube	AI recommends videos based on your viewing behavior.	Your homepage shows videos related to your interests.
7	Email	AI identifies unwanted or suspicious emails.	Gmail automatically moves many spam emails into the Spam folder.
8	Online Shopping	AI studies your searches and purchases.	Amazon or other shopping apps recommend products you may want to buy.
9	Face Unlock	AI can recognize a person's face.	A smartphone unlocks when it recognizes the owner's face.
10	Social Media	AI recommends posts, reels, and content based on user activity.	Instagram shows content similar to what you frequently watch or like.
11	Online Banking	AI can help detect unusual or suspicious transactions.	A bank may flag a transaction that looks different from your normal activity.
12	Chatbots	AI understands questions and generates responses.	Customer-support chatbots answer common customer questions.

How AI Works in These Applications

Most of these applications follow a simple process:



Example: Spotify



Machine Learning Concepts and Learning Strategies

Machine Learning (ML) is a part of Artificial Intelligence that allows computers to **learn from data and make predictions or decisions without being explicitly programmed for every situation.**

There are three main learning strategies:

1. **Supervised Learning**
2. **Unsupervised Learning**
3. **Reinforcement Learning**

1. Supervised Learning

In **Supervised Learning**, the machine learns from **labeled data**.

Here, we provide the machine with:

- **Input data**
- **Correct output**

The machine learns the relationship between them and then predicts the output for new data.

Simple Example

Suppose we want a computer to identify whether an email is **Spam or Not Spam**.

We give it many examples:

Email	Correct Answer
"You won a lottery!"	Spam
"Get free money now!"	Spam
"Meeting at 10 AM"	Not Spam

Email	Correct Answer
"Your assignment is ready"	Not Spam

The machine learns from these examples.

When a new email arrives, it predicts whether it is **Spam or Not Spam**.

Common Applications

- Spam email detection
- House price prediction
- Student result prediction
- Disease classification

2. Unsupervised Learning

In **Unsupervised Learning**, the machine learns from **unlabeled data**.

We give the machine data, but **we do not provide the correct answers**.

The machine tries to find:

- Groups
- Similarities
- Patterns
- Relationships

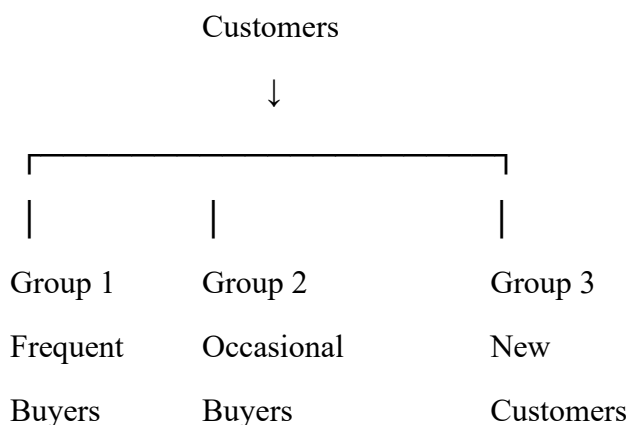
by itself.

Example

Suppose a shopping website has information about thousands of customers.

We don't tell the machine which customers belong to which group.

The machine may find groups such as:



The groups are created based on similarities in customer behavior.

Common Applications

- Customer grouping
- Market segmentation
- Finding similar documents
- Recommendation systems
- Detecting unusual patterns

3. Reinforcement Learning

In **Reinforcement Learning**, a machine or **agent learns by interacting with its environment**.

The machine:

1. Takes an action.
2. Gets a **reward or penalty**.
3. Learns from the result.
4. Tries to improve its future actions.

Simple Example

Imagine teaching a robot to reach a destination.

Robot takes correct action



Reward (+)



Robot learns that action

If it takes a wrong action:

Robot takes wrong action



Penalty (-)



Robot tries another action

After many attempts, the robot learns which actions give better results.

Common Applications

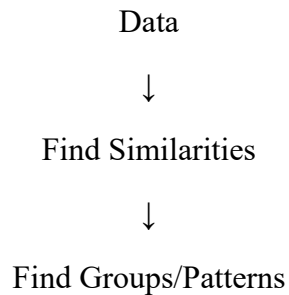
- Game playing
- Robotics
- Self-driving systems
- Resource management
- Recommendation and decision systems

Difference Between Supervised, Unsupervised and Reinforcement Learning

Point	Supervised Learning	Unsupervised Learning	Reinforcement Learning
1.Learning Type	Learns from labeled data.	Learns from unlabeled data.	Learns through interaction.
2.Correct Answer	Correct answer is provided during training.	Correct answer is not provided.	No fixed answer is provided; rewards guide learning.
3.Main Purpose	Make predictions or classifications.	Find hidden patterns or groups.	Find the best actions to achieve a goal.
4. Feedback	Gets feedback through known correct outputs.	Usually no direct feedback.	Gets rewards or penalties.
5. Data	Labeled data.	Unlabeled data.	Experience from actions and outcomes.
6.Learning Method	Learns input-output relationships.	Finds similarities and patterns.	Learns which actions produce better results.
7. Example	Spam detection.	Customer grouping.	Teaching a robot to move.
8. Output	Known categories or predicted values.	Groups, patterns, or relationships.	Actions or decisions.
9. Human Role	Humans provide labeled training data.	Humans provide data but generally not labels.	Humans define the environment, actions, and reward system.
10.Common Algorithms	Linear Regression, Decision Tree, SVM, etc.	K-Means, Hierarchical Clustering, etc.	Q-Learning, Deep Q-Networks, etc.

Unsupervised Learning

Student gets data but no answers

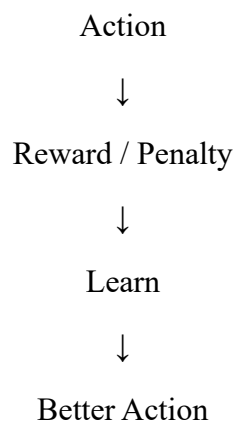


Example:

Machine receives many animal pictures and tries to group similar pictures together without being told their names.

Reinforcement Learning

Learn by trial and error



Example:

A computer learns to play a game by trying different moves and receiving rewards for successful moves.

Data, Data Visualization and Data Pre-processing

These are important concepts in **Machine Learning** because a machine learns from data. Before giving data to a machine-learning model, it is often necessary to understand, clean, and prepare the data.

1. Data

Data is a collection of information that can be used for analysis and decision-making.

Data can be in the form of:

- Numbers
- Text

- Images
- Audio
- Video
- Tables

Example

Student data:

Name	Age	Attendance	Marks
Rahul	18	85%	78
Priya	19	92%	88
Aman	18	70%	65

Here, **Name**, **Age**, **Attendance**, and **Marks** are examples of data.

Types of Data

Type	Meaning	Example
Structured Data	Data arranged in a fixed format such as rows and columns.	Student table
Unstructured Data	Data without a fixed structure.	Images, videos, audio
Semi-structured Data	Data that has some organization but not a fixed table format.	JSON, XML

In Machine Learning

Data is used to:

1. Train the machine-learning model.
2. Test the model.
3. Make predictions on new information.

2. Data Visualization

Data Visualization is the process of representing data graphically using graphs, charts, or diagrams so that it becomes easier to understand.

Instead of looking at a large table, we can use a graph to quickly understand the information.

Common Types

Visualization	Used For
Bar Chart	Comparing different categories
Line Chart	Showing changes over time
Pie Chart	Showing parts of a whole
Histogram	Showing distribution of numerical data
Scatter Plot	Showing the relationship between two variables

Example

Suppose the marks of students are:

Rahul → 78

Priya → 88

Aman → 65

Neha → 92

A **bar chart** can be used to easily compare their marks.

Why Data Visualization is Used?

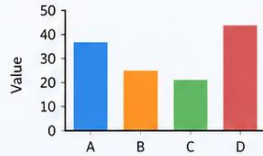
1. Makes data easier to understand.
2. Helps identify patterns.
3. Helps identify trends.
4. Helps find unusual values.
5. Makes comparison easier.
6. Helps in making better decisions.

Types of Charts in Data Visualization

(with simple examples)

1. Bar Chart

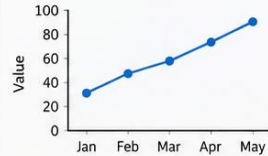
Used to compare different categories.



Example: Sales of different products

2. Line Chart

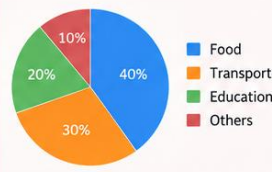
Used to show changes over time.



Example: Monthly temperature

3. Pie Chart

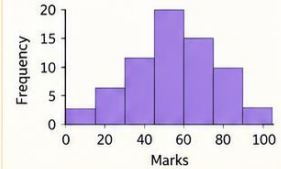
Used to show parts of a whole.



Example: Monthly expenses

4. Histogram

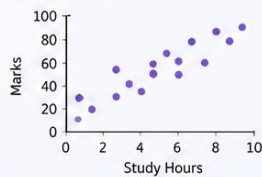
Used to show the distribution of numerical data.



Example: Distribution of student marks

5. Scatter Plot

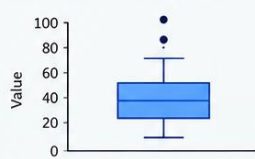
Used to show the relationship between two numerical variables.



Example: Study hours vs Marks

6. Box Plot

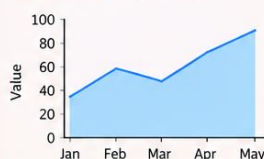
Used to show the spread of data (median, quartiles, outliers).



Example: Marks of students

7. Area Chart

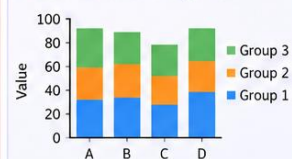
Used to show change in quantity over time with filled area.



Example: Website visitors over months

8. Stacked Bar Chart

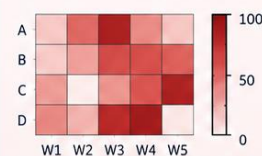
Used to show parts of a whole across different categories.



Example: Sales by category and region

9. Heatmap

Used to show data values using colors (good for large data).



Example: Performance of students across weeks

10. Treemap

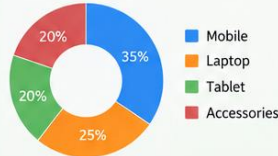
Used to show hierarchical data using nested rectangles.



Example: Product categories in a store

11. Donut Chart

Similar to pie chart but with a hole in the center.



Example: Device usage

12. Radar Chart

Used to compare multiple variables.



Example: Student performance in subjects

Different charts are used for different types of data and purposes.
Choose the right chart to make your data easy to understand!

3. Data Pre-processing

Data Pre-processing is the process of **cleaning and preparing raw data before using it for Machine Learning**.

Raw data may contain:

- Missing values
- Incorrect values
- Duplicate data
- Different formats
- Unwanted information

Therefore, data needs to be prepared before giving it to the ML model.

Example

Suppose we have:

Student	Age	Marks
Rahul	18	78
Priya	Missing	88
Aman	18	-5
Rahul	18	78

Here:

- Age of Priya is missing.
- Aman has an incorrect marks value.
- Rahul's record is duplicated.

These problems should be handled before training the model.

Main Steps of Data Pre-processing

1. Data Collection

Collect the required data from different sources.

Example:

Student information collected from forms, databases, or files.

2. Data Cleaning

Remove or correct problems present in the raw data.

It includes:

- **Handling Missing Values: -**

Missing values can be:

- Removed
- Replaced with a suitable value
- Filled using other methods

Example:

Age = 18, 19, Missing, 20

The missing value can be filled appropriately.

- **Data Scaling: -**

Bringing numerical values into a suitable range when required.

Example:

Age: 18, 20, 22

Salary: 25000, 50000, 75000

These values have very different scales, so scaling may be useful for some ML algorithms.

- **Removing duplicate records: -**

Repeated records are identified and removed when they are not needed.

- **Handling Outliers: -**

An **outlier** is a value that is very different from most other values.

Example:

10, 12, 11, 13, 12, 150

Here, **150** may be an outlier depending on the context.

- Correcting incorrect data
- Handling inconsistent data
- Handling outliers

Example:

Age: 18, 19, Missing, 20

The missing value needs to be handled before using the data.

3. Data Integration

Combine data collected from different sources into a single dataset.

Example:

Student Database + Attendance Database

↓

Combined Dataset

4. Data Transformation

Convert data into a suitable format for analysis or machine learning.

It includes:

- Normalization
- Standardization

- Encoding categorical data
- Converting data into suitable formats

Example:

Gender:

Male → 0

Female → 1

5. Data Reduction

Reduce the size or complexity of data while keeping the important information.

It can include:

- Removing unnecessary features
- Selecting important data
- Reducing the number of records when appropriate

Example:

If a dataset has 50 columns but only 10 are useful for prediction, unnecessary columns can be removed.

6. Data Splitting

Divide the prepared data into different parts for machine learning.

Commonly:

- Training data: Used to train the model.
- Testing data: Used to check the model's performance.

Main Step	What is done?
1. Data Collection	Collect the required data.
2. Data Cleaning	Remove or correct errors, missing values, duplicates, etc.
3. Data Integration	Combine data from different sources.
4. Data Transformation	Convert data into a suitable format.
5. Data Reduction	Remove unnecessary data and reduce complexity.
6. Data Splitting	Divide data into training and testing sets.

Difference Between Data, Data Visualization and Data Pre-processing

Point	Data	Data Visualization	Data Pre-processing
1. Meaning	Collection of information.	Graphical representation of data.	Process of preparing data for use.
2. Purpose	Provides information for analysis.	Makes information easier to understand.	Makes data suitable for ML models.
3. Form	Numbers, text, images, etc.	Charts, graphs, and diagrams.	Cleaned and transformed data.
4. Example	Student marks.	Bar chart of student marks.	Removing missing or incorrect marks.
5. Main Activity	Collecting information.	Representing information visually.	Cleaning and transforming information.
6. Used Before ML?	Yes.	Helps understand the data.	Yes, commonly before training an ML model.

Unit - 2

Machine Learning: Classification Technique — K-Nearest Neighbors (KNN)

1. What is Classification?

Classification is a supervised machine learning technique used to **divide data into different predefined categories or classes**.

Examples:

- Email → **Spam / Not Spam**
- Student → **Pass / Fail**
- Disease → **Yes / No**
- Image → **Cat / Dog**

2. What is K-Nearest Neighbors (KNN)?

K-Nearest Neighbors (KNN) is a simple **supervised machine learning classification algorithm**.

It classifies a new data point by looking at the **K nearest data points** that are already known.

Simple idea:

A new data point usually belongs to the class of its nearest neighbors.

KNN does not build a complicated model during training. Instead, it stores the existing data and compares a new data point with it when a prediction is required.

3. How KNN Works

Suppose we have two classes:

-  Class A
-  Class B

For a new point, KNN follows these steps:

Step 1: Choose K

Select the number of neighbors, such as: **K = 3**

Step 2: Calculate Distance

Calculate the distance between the new point and existing data points.

The commonly used distance is **Euclidean Distance**.

Step 3: Find K Nearest Points

Select the **K points having the smallest distances**.

Step 4: Voting

Check the classes of those K nearest points.

Step 5: Assign the Class

The class having the **maximum number of votes** becomes the class of the new data point.

4. KNN Example

Suppose we want to classify a student based on:

- **Study Hours**
- **Marks**

We have the following data:

Student	Study Hours (X)	Marks (Y)	Class
A	2	50	Fail
B	3	60	Fail
C	4	70	Pass
D	5	80	Pass
E	6	90	Pass

New student:

P = (4, 75)

Let: **K = 3**

We calculate the distance of P from every student.

5. Euclidean Distance

The formula for two points:

$$\begin{array}{l} (x_1, y_1) \\ \text{and} \\ (x_2, y_2) \\ \text{is:} \\ d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2} \end{array}$$

Find the difference between the X values and Y values, square them, add them, and take the square root.

6. Numerical Problem

Given:

New point:

P = (4,75)

Existing points:

- A = (2,50) → Fail
- B = (3,60) → Fail
- C = (4,70) → Pass
- D = (5,80) → Pass
- E = (6,90) → Pass

We need to find the **3 nearest neighbors**.

Distance from A

$$\begin{aligned}d_A &= \sqrt{(4 - 2)^2 + (75 - 50)^2} \\&= \sqrt{2^2 + 25^2} \\&= \sqrt{4 + 625} \\&= \sqrt{629} \\d_A &\approx 25.08\end{aligned}$$

Distance from B

$$\begin{aligned}d_B &= \sqrt{(4 - 3)^2 + (75 - 60)^2} \\&= \sqrt{1^2 + 15^2} \\&= \sqrt{226} \\d_B &\approx 15.03\end{aligned}$$

Distance from C

$$\begin{aligned}d_C &= \sqrt{(4 - 4)^2 + (75 - 70)^2} \\&= \sqrt{0 + 25} \\d_C &= 5\end{aligned}$$

Distance from D

$$\begin{aligned}d_D &= \sqrt{(4 - 5)^2 + (75 - 80)^2} \\&= \sqrt{1 + 25} \\&= \sqrt{26} \\d_D &\approx 5.10\end{aligned}$$

Distance from E

$$\begin{aligned}d_E &= \sqrt{(4 - 6)^2 + (75 - 90)^2} \\&= \sqrt{4 + 225} \\&= \sqrt{229} \\d_E &\approx 15.13\end{aligned}$$

7. Arrange Distances

Student	Distance	Class
C	5.00	Pass
D	5.10	Pass
B	15.03	Fail
E	15.13	Pass
A	25.08	Fail

Since:

$$K = 3$$

we select the **3 nearest neighbors**:

1. $C \rightarrow \text{Pass}$
2. $D \rightarrow \text{Pass}$
3. $B \rightarrow \text{Fail}$

Voting

Class	Votes
Pass	2
Fail	1

Therefore:

New Student = Pass

8. KNN Algorithm

New Data Point



Select value of K



Calculate distances



Find K nearest points



Check their classes



Majority voting



Predicted Class

Example: Classifying a Student

Suppose we have students based on their **study hours** and **marks**.

Student	Study Hours	Marks	Result
A	2	40	Fail
B	3	50	Fail
C	4	70	Pass
D	5	80	Pass
E	6	90	Pass

Now we have a **new student X**:

Study Hours = 4.5

Marks = 75

We want to predict whether the student will **Pass or Fail**.

Step 1: Choose K

Suppose: **K = 3**

This means KNN will look at the **3 nearest students**.

Step 2: Find the nearest students

After comparing the new student with the existing students, suppose the 3 nearest students are:

Neighbor	Result
C	Pass
D	Pass
B	Fail

Step 3: Voting

Now count the results:

Pass → 2, Fail → 1 vote

Pass has more votes.

Therefore:

Prediction = PASS